TASK 4: MEDIA PRODUCTION COST TRACKING

SQL • Google Sheets / Excel • Cost analysis and ROI examples

1. Objective

Track production costs for digital media assets and compare formats such as DSLR portrait sets and cinematic animation videos. Calculate ROI only if a reliable attributed-revenue or defined value field is available.

2. Main calculations

- Total cost = pre-production + production + editing + distribution costs.
- Net return = attributed revenue – total cost.
- ROI (%) = (net return ÷ total cost) × 100, when total cost is greater than zero.
- Cost per conversion = total cost ÷ conversions, if conversion counts are provided.

3. Synthetic example data

Asset ID	Format	Pre-prod	Production	Editing	Distribution	Revenue
A01	Portrait	5000	12000	3000	2000	35000
A02	Animation	8000	22000	9000	4000	70000

Example A01: total cost = ■5,000 + ■12,000 + ■3,000 + ■2,000 = ■22,000. Net return = ■35,000 – ■22,000 = ■13,000. ROI = ■13,000 / ■22,000 × 100 = 59.1%. All sample values are fictional.

4. Spreadsheet setup

Example columns: A Asset ID, B Format, C Pre-production, D Production, E Editing, F Distribution, G Attributed Revenue, H Total Cost, I Net Return, J ROI.

```
H2: =SUM(C2:F2)
I2: =G2-H2
J2: =IF(H2>0,I2/H2,"")
Format column J as Percentage.
```

Use numeric cells for money, not text. Decide whether taxes, overhead, staff time, and agency fees are included, then apply the same rules consistently.

5. SQL table

```
CREATE TABLE media_assets (
  asset_id VARCHAR(20) PRIMARY KEY,
  asset_name VARCHAR(120),
  format VARCHAR(40),
  production_date DATE,
  preproduction_cost NUMERIC(12,2),
  production_cost NUMERIC(12,2),
  editing_cost NUMERIC(12,2),
  distribution_cost NUMERIC(12,2),
  revenue_attributed NUMERIC(12,2)
);
```

6. SQL summary by media format

```
WITH totals AS (
  SELECT format,
```

```
    COALESCE(preproduction_cost,0) + COALESCE(production_cost,0)
      + COALESCE(editing_cost,0) + COALESCE(distribution_cost,0) AS total_cost,
    revenue_attributed
  FROM media_assets
)
SELECT format,
       COUNT(*) AS asset_count,
       ROUND(AVG(total_cost), 2) AS avg_cost,
       ROUND(SUM(total_cost), 2) AS total_spend,
       ROUND(SUM(revenue_attributed), 2) AS attributed_revenue,
       ROUND(100.0 * SUM(revenue_attributed - total_cost)
             / NULLIF(SUM(total_cost),0), 2) AS roi_percent
FROM totals
GROUP BY format
ORDER BY total_spend DESC;
```

This example uses PostgreSQL-style syntax. MySQL may require small changes. COALESCE treats a missing cost as zero, which is appropriate only if that is a valid rule; otherwise investigate missing values.

7. Data quality and interpretation

- Check duplicate asset IDs, negative costs, missing values, and inconsistent format labels.
- Use comparable attribution windows and the same definition of revenue across formats.
- Portfolio ROI is not the same as the average of individual asset ROI percentages.
- Example conclusion: “Animation has higher average cost and higher attributed revenue in this sample. Compare cost per conversion before reallocating budget.”

Task 4 deliverables

- Spreadsheet with formulas and documented assumptions.
- SQL table/query script.
- Summary by media format: count, average cost, total spend, revenue, ROI.
- Short conclusions and limitations.